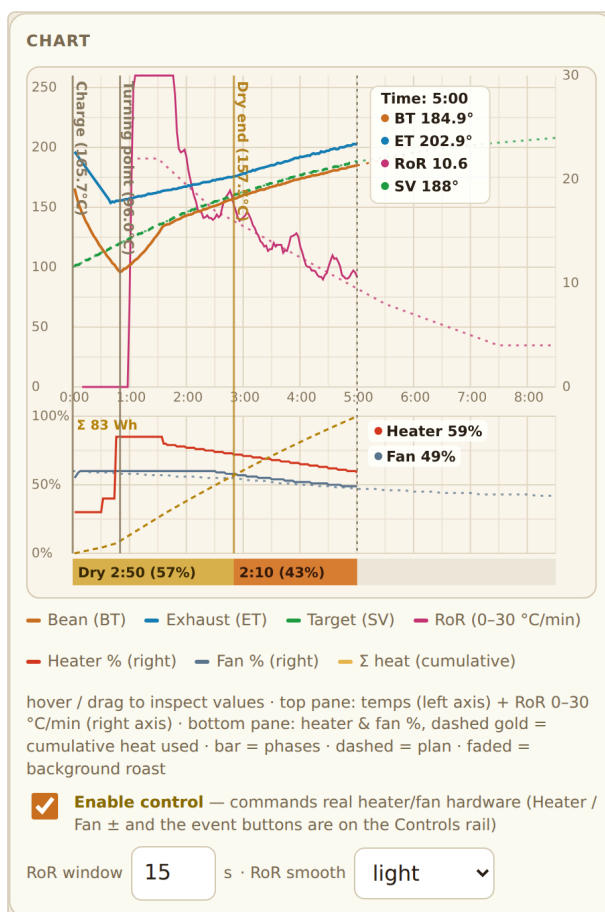


Roast Controller

User Manual

Web-Bluetooth controller for Hamid / Matchbox (TC4-class) coffee roasters



Covers setup, your first roast, the everyday roasting loop, profile design, inventory, cloud sync and troubleshooting. Screenshots show the app in day mode. Edition of September 2026.

Contents

- | | |
|--------------------------------------|---|
| 1. Before you start | 8. After the roast: data, notes and comparisons |
| 2. Getting to know the screen | 9. Green coffee inventory |
| 3. First-time setup | 10. Using more than one device (Cloud sync) |
| 4. Your first roast, step by step | 11. Advanced mode |
| 5. The everyday workflow | 12. Safety checklist |
| 6. Building your own profile | 13. Troubleshooting & FAQ |
| 7. Reading the screen during a roast | |

Read this first. The app drives a real heating element. Controls stay locked until you tick **Enable control**. Keep the fan running whenever the heater is on, never leave a roast unattended, and press Stop / Drop (the red button on the Controls rail) if anything looks wrong. It always works, whatever else is running.

1. Before you start

You need

- **Chrome or Edge** on Android, Windows, macOS, Linux or ChromeOS, with Bluetooth switched on. iOS Safari does not support Web Bluetooth, so an iPhone or iPad will not work. A desktop or an Android phone or tablet is the usual choice.
- The app opened over **HTTPS** (the hosted link you were given). Opening the file from disk will not work because Web Bluetooth needs a secure page.
- A roaster whose Bluetooth module advertises as **MATCHBOX** or **MBOX**, powered on and within a few metres.
- Green coffee, a scale, and somewhere ventilated to roast.

Good to know

- Everything you save (profiles, roast logs, inventory, settings) is stored in **this browser on this device**. Nothing leaves your machine unless you switch on Cloud sync (section 10).
- The 🌞 / 🌙 button in the header switches day and night mode. The choice is remembered.
- The page is one long column of panels. This manual follows them from top to bottom.

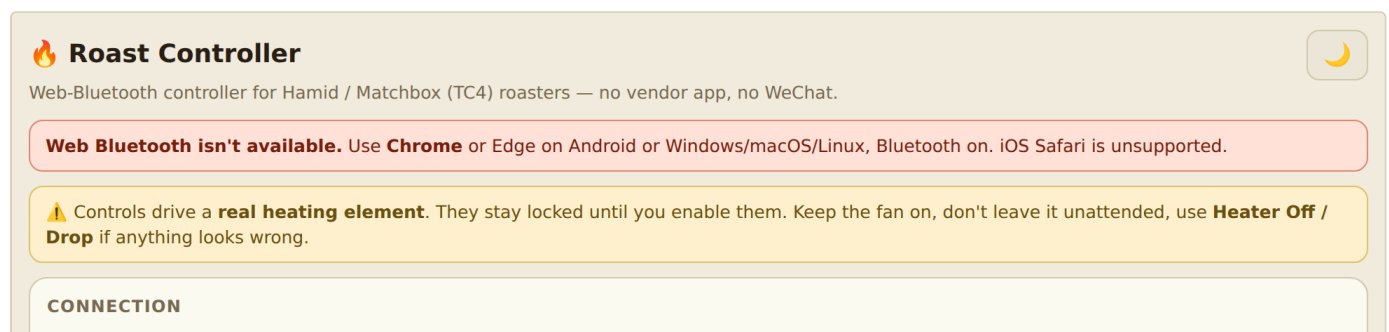


Fig. 1 — The header with the day/night toggle, and the safety banner shown on every visit.

2. Getting to know the screen

The whole app is one page. Scroll down and you pass through these panels in order:



Fig. 2 — The full page at a glance (day mode). The Live rail sits beside the chart on wide screens and above it on phones.

1. **Connection** — connect or disconnect the roaster, see the Bluetooth details.
2. **Live** — roast timer, batch code, the six live tiles (Bean, ET, RoR, Target, Heater, Fan), the previous event and live phase bar.
3. **Chart** — temperatures, RoR, heater and fan over time, the **Enable control** tick and the RoR settings.
4. **Controls** rail — Target temp and PID tiles, Heater and Fan $\pm$ steppers, event tagging and **Stop / Drop**. This is where you drive the roast.
5. **Between batches / charge** — cool-to, charge temperature and fan, the charge soak, Auto-charge, and the **Start between-batch cycle** button.
6. **Profile designer** — pick, generate, draw, import, export and run roast profiles.
7. **Advanced mode** tick — reveals the ET-track and Power profile tools (section 11).
8. **Data** — CSV export, saved roasts, roast notes and stars, background overlay, raw log.
9. **Cloud sync** — share your data between devices through your own Cloudflare Worker.
10. **Green coffee inventory** — beans in stock, label photo reader, batch planner.

3. First-time setup (5 minutes)

1. Power the roaster on. Close the vendor app if it is open. Only one thing can hold the Bluetooth connection.
2. Open the app and press **Connect roaster** in the **Connection** panel. A browser dialog lists nearby devices. Pick **MATCHBOX** or **MBOX**. If it is not listed, tick **Can't find it by name? List all Bluetooth devices** and try again.
3. The status pill turns to **connected** and the **Live** tiles start showing Bean °C, ET °C, Heater % and Fan %. If the numbers appear, the link is working.
4. Tick **Enable control** just under the chart. Until this is ticked the app only *reads* from the roaster.
5. Test the fan: tap **Fan +** on the **Controls** rail a few times. You should hear the roaster's fan change and see the Fan value climb. This confirms commands reach the machine.

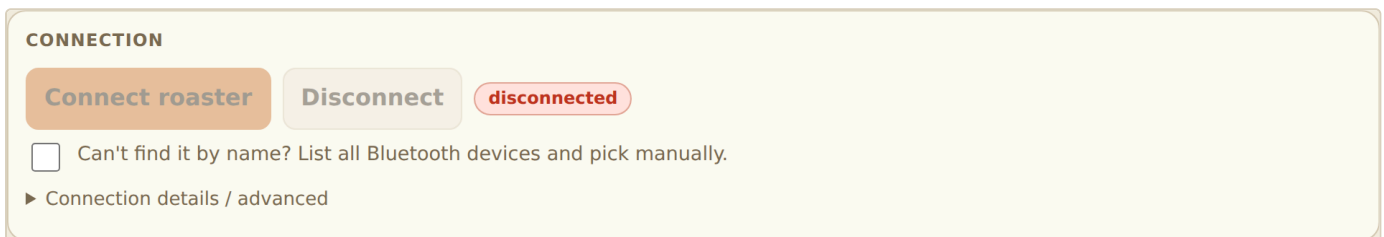


Fig. 3 — The Connection panel before connecting.

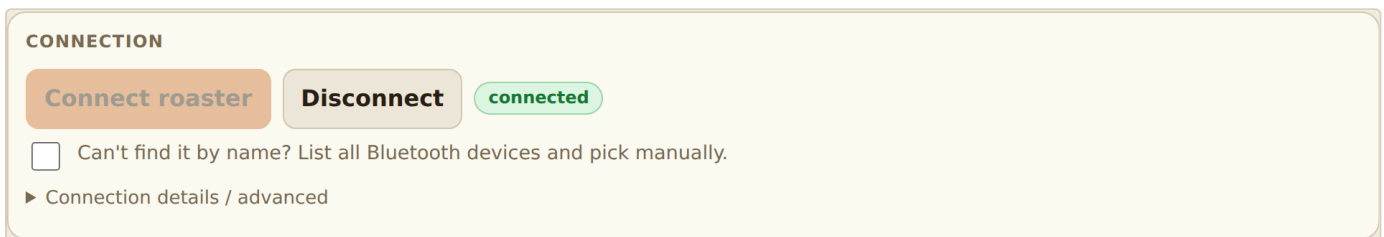


Fig. 4 — Connected. The pill turns green and Disconnect becomes available.

If the tiles stay at – after connecting, open **Connection details / advanced** and see the troubleshooting table in section 13.

4. Your first roast, step by step

The app can run the *whole* roast for you from a single button. For the first roast, use a ready-made profile so you can learn the screen without designing a curve.

A. Load a profile

1. Download one of the ready-made curves from the repository's `profiles/` folder. `sprint-140-filter-208.json` is a good all-round filter roast for about 150 g.
2. In the **Profile designer** panel press `Import JSON` and choose the file. The curve appears on the graph with the name and notes filled in.
3. Press `Save profile`. It now appears in the profile drop-down for next time.
4. Check that **Auto fan** below the graph is ticked (the sample profiles set it). Leave the rest as is.

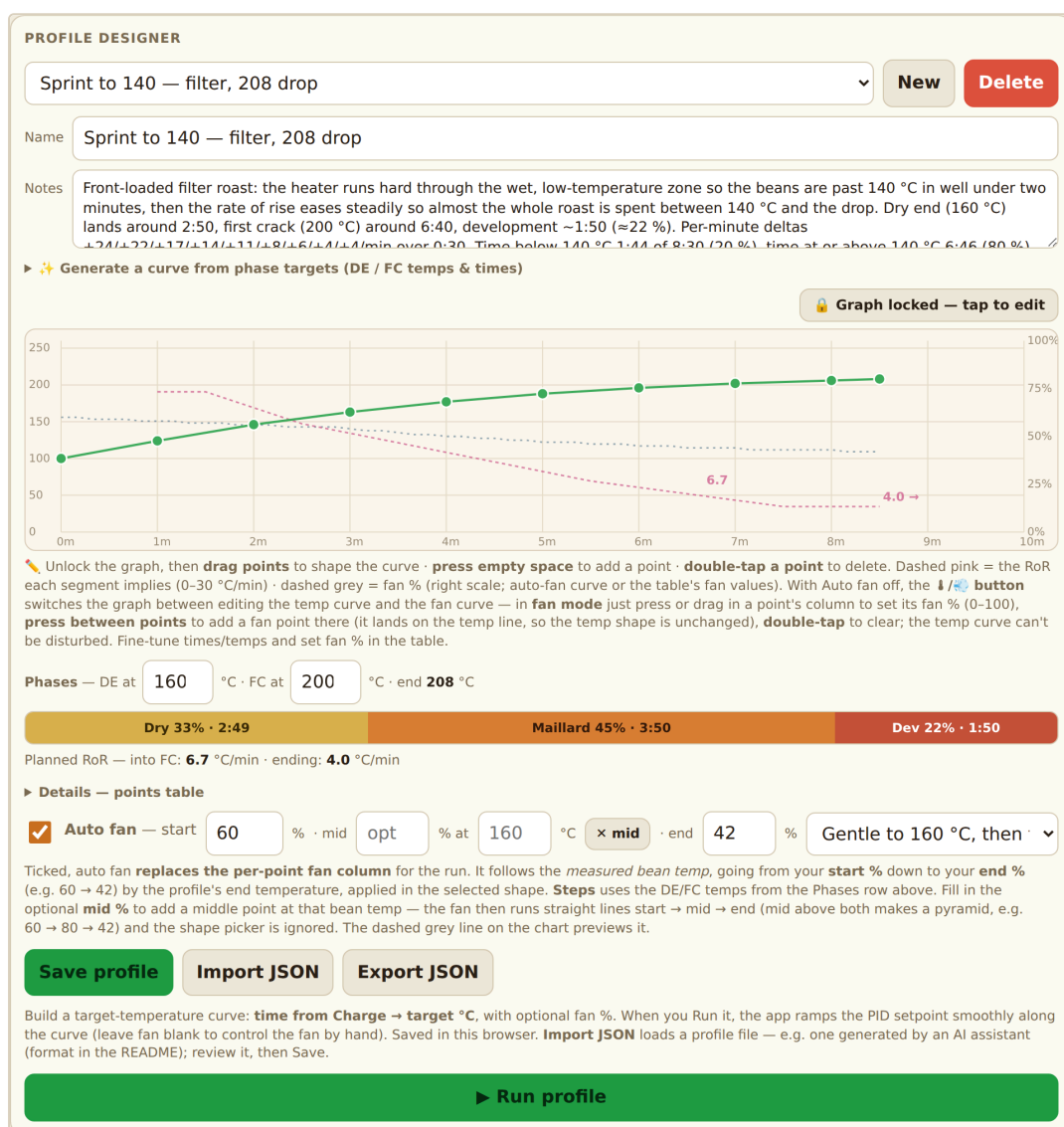


Fig. 5 — The sample profile loaded and saved. Green line: target bean temperature. Dashed pink: the RoR each segment implies. Dashed grey: the auto-fan curve. The phase bar shows the Dry / Maillard / Development split.

B. Set the between-batch cycle

In the **Between batches / charge** panel:

- **Cool to °C** — the machine is fanned down to this before charging (default 100).
- **Charge temp °C** — the drum temperature at which you drop beans in (default 180). Start with the default.
- **Charge fan %** — fan speed used while heating to charge and at the moment you pour (default 30).
- Leave **Charge soak** and **Auto-charge** ticked. Under **At bean drop** leave **Run the profile** selected. The preview box confirms which profile will run when the beans go in.

BETWEEN BATCHES / CHARGE

Start between-batch cycle

Cools the machine (fan 100%, heater off) to the cool-to temp, then ramps gradually to your charge temp over **1 min** at the charge fan speed, holds it **30 s**, and lights up when it's time to drop the beans.

At bean drop ☒ **Run the profile** (or an armed replay) ☐ **Manual** — hand control from the Controls rail

At bean drop this will run:
Profile — Sprint to 140 — filter, 208 drop
 8:30 → 208 °C · auto fan 60 → 42% (knee)

Cool to °C Charge temp °C Charge fan %

☒ **Charge soak** — heater % until s, then % until s

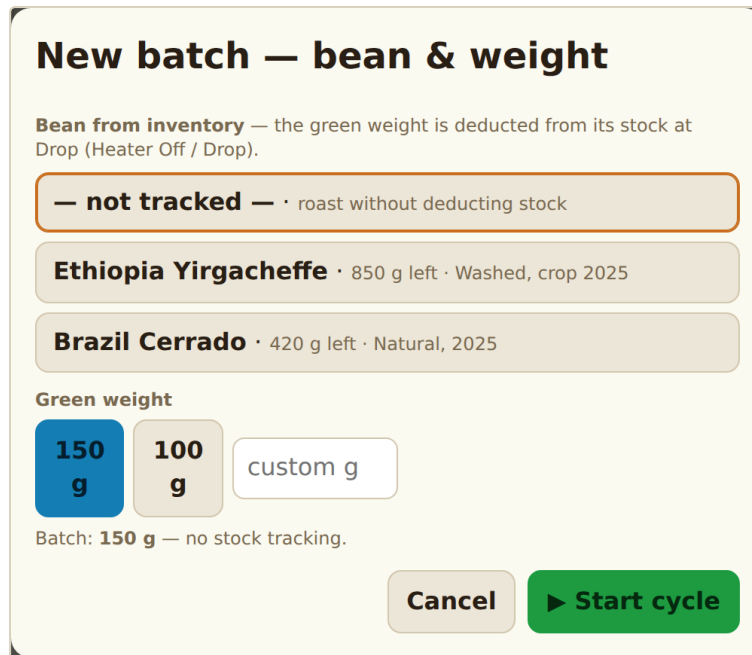
Fixed heater power right after **Charge** (open loop, PID off) — then the control loop takes over. Applies to profile runs **and ET-track runs** started at charge. **Untick to skip the soak** — the control loop then engages on the curve from the first second after charge. Saved in this browser. ⚡ Charging and the between-batch cycle are **hard-capped at 85% power** — above that the app pauses the PID and holds 85% until the temperature is nearly there.

☒ **Auto-charge** — when the beans drop (BT plunges), mark Charge and start the profile loaded in the designer automatically. **No need to press ► Run profile** — the whole roast runs from this one button.

Fig. 6 — Between batches / charge, with the defaults. The grey box previews what will run at bean drop.

C. Run the cycle

1. Press **Start between-batch cycle**. A dialog asks for the **bean** (pick one from inventory, or leave it untracked) and the **green weight** (150 g, 100 g, or a custom figure). Both are stored with the roast, and the green weight is taken out of that bean's stock at Drop. Press **Start cycle**.
2. The app fans the machine down to the cool-to temperature, then ramps to the charge temperature over about a minute and holds it for 30 s.
3. When the drum is stable the panel flashes **↓ DROP BEANS NOW** and the device beeps or vibrates. Pour the beans in.
4. With **Auto-charge** on, the app sees the bean-probe plunge, marks **Charge**, starts the timer, and begins the profile. You do not need to press Run profile.
5. For the first 45 s the heater runs at a fixed low power (the *charge soak*), then the PID takes over and follows your curve.



New batch — bean & weight

Bean from inventory — the green weight is deducted from its stock at Drop (Heater Off / Drop).

— not tracked — · roast without deducting stock

Ethiopia Yirgacheffe · 850 g left · Washed, crop 2025

Brazil Cerrado · 420 g left · Natural, 2025

Green weight

150 g 100 g custom g

Batch: 150 g — no stock tracking.

Cancel Start cycle

Fig. 7 — The New batch dialog: choose the bean and the green weight, then Start cycle.

D. During the roast

- Watch **Bean °C** climb and the **RoR** tile settle. On the chart the plan is dashed and the actual bean temperature is solid.
- Press the green **DE** button on the Controls rail when the beans turn from green-yellow to tan (around 150–165 °C bean temperature). It then becomes **FC**, which you press when you hear the first pops. These marks drive the phase bar and the development-time percentage.
- If the beans are not tumbling well, bump the fan with **Fan +** on the rail. This pauses the automated fan. **Resume profile fan** under the chart hands it back.

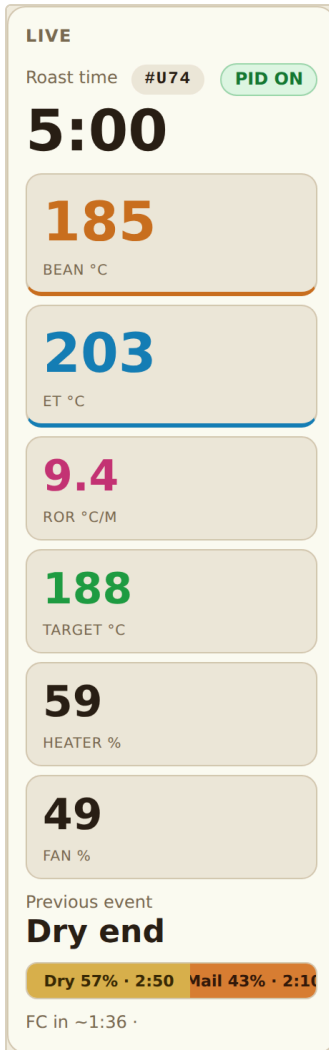


Fig. 8 — The Live rail at 5:00: timer, batch code, tiles, previous event, phase bar and the FC countdown.

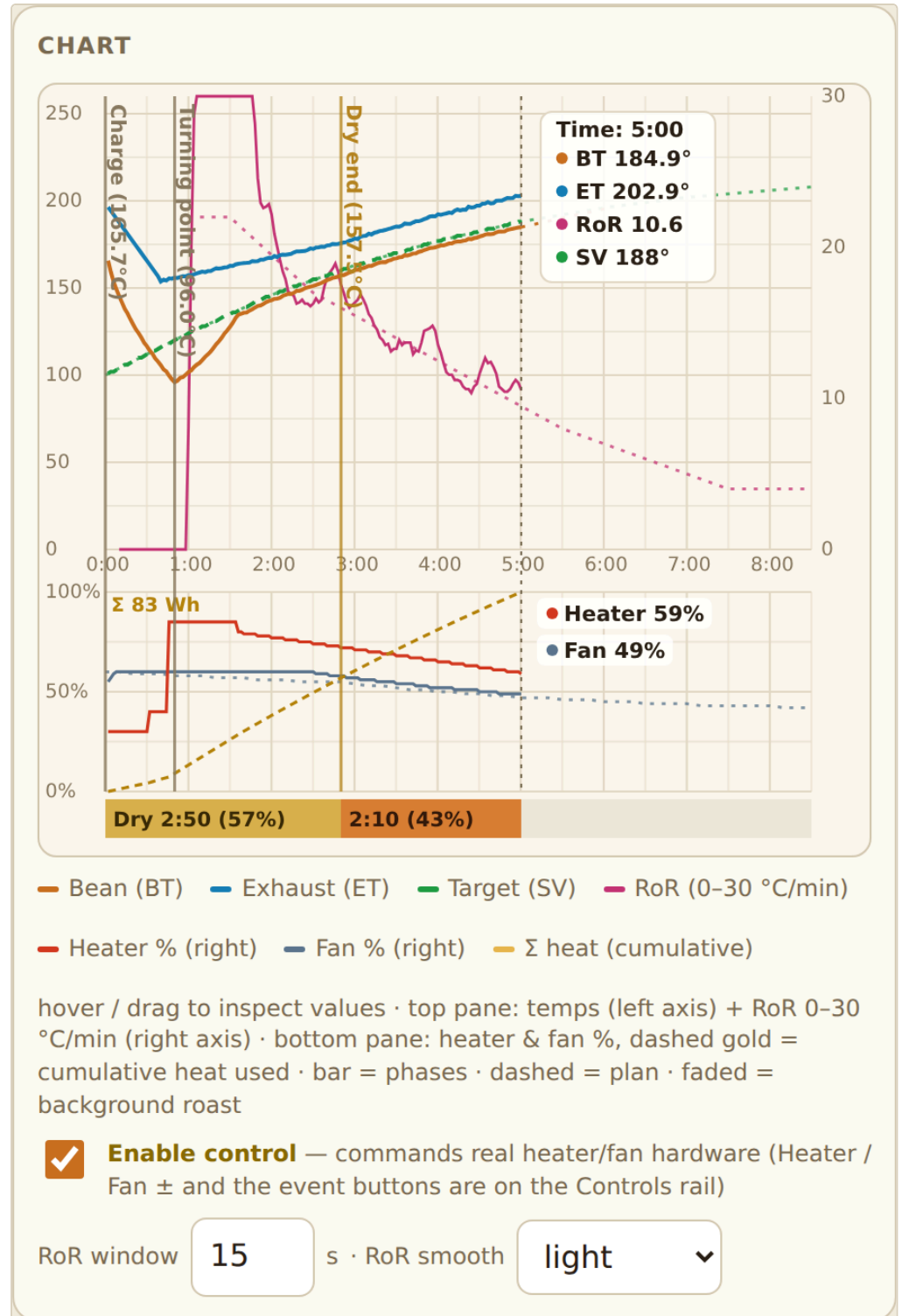


Fig. 9 — The chart at 5:00 with Charge, turning point and Dry End marked. Hover or drag to read values. Below the chart: the Enable control tick and the RoR settings.

E. Drop

1. When the beans reach the colour you want, press **Stop / Drop** on the Controls rail. The heater goes to 0, the fan returns to the charge speed, the roast is stamped with a 3-character **batch code** (for example #K7Q) and is **saved automatically**. Write the code on the bag.
2. If you let the profile run to its last point instead, the heater switches off by itself and an alert sounds. You still press **Stop / Drop** to end and save the roast.

3. The saved roast appears at the top of the saved-roasts list in the **Data** panel, named by its batch code, time and bean. Type tasting notes there whenever you like.
4. Tip the beans out to cool and let the machine idle with the fan on before the next batch.

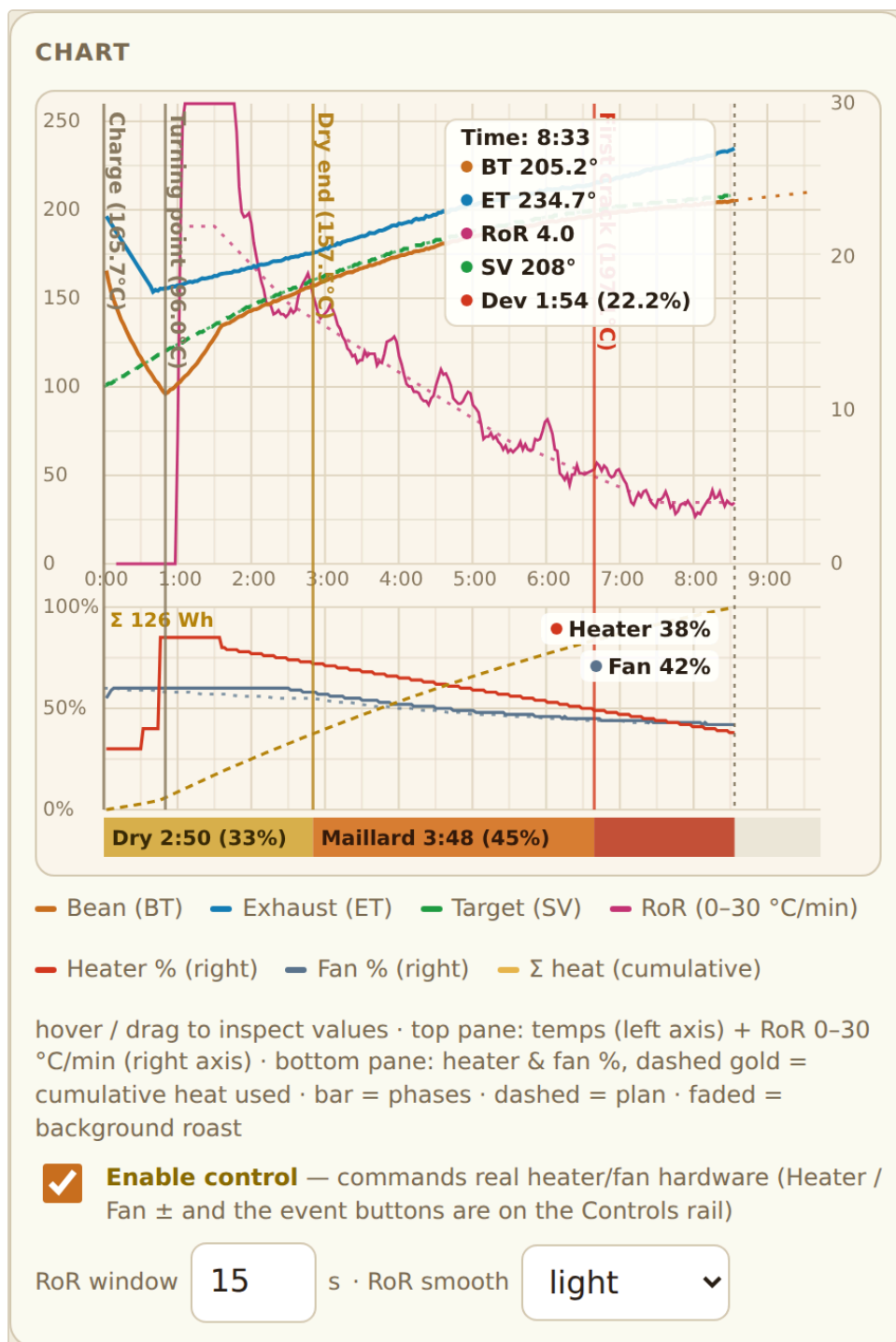


Fig. 10 — Just before Drop at 8:30. First Crack is marked and the development phase is shown on the bar.

That is the whole loop. The next batch starts again at step C.

5. The everyday workflow

Connect → Enable control → pick profile → Start between-batch cycle
→ DROP BEANS NOW → (auto-charge) → mark Dry End / First Crack
→ Stop / Drop → notes / star → next batch

Between batches the app does the cooling and re-heating for you, so the rhythm is: *tip out, note, press Start cycle, wait for the light, pour.*

Tips for a smooth session

- Set the profile and the between-batch values once at the start. They are remembered.
- Keep the bag or a notebook beside you: the batch code is shown in the Live rail from Charge onward.
- Use **Overlay as background** (section 8) with a roast you liked, and roast the next batch to match it.
- Green weight is deducted from inventory at Drop, so keep your stock up to date and the batch planner will keep suggesting even splits.

6. Building your own profile

Open the **Profile designer** panel. A profile is a target bean-temperature curve over time (from Charge), plus an optional fan schedule.

Fastest: generate from phase targets

1. Open  **Generate a curve from phase targets**.
2. Enter the temperature you want at 0:00, your **dry-end** temperature and how long drying should take, your **first-crack** temperature and how long the Maillard phase should take, and the **drop** temperature and development time.
3. Pick a **shape**. If unsure, start with **Declining RoR** (the classic). The others are explained in the panel text. **Sprint** and **Flat development** are popular next steps.
4. Optionally tick  **Precharge** (extra push in the first 40 s) or  **Coast** (finish 3 °C early on stored heat).
5. The curve loads into the graph with the name, notes and phase markers filled in. Flip between shapes to compare, then Save profile.

 **Generate a curve from phase targets (DE / FC temps & times)**

Start	100	°C target at 0:00 (right after Charge)
Dry end	160	°C, drying takes 4 min 0 sec
First crack	200	°C, Maillard takes 3 min 30 sec
Drop	210	°C, development takes 2 min 0 sec

 **Declining RoR**

 **Pyramid RoR**

 **Straight phases**

 **Soak start**

 **Flat development**

 **Step-down at FC**

 **Boil-off plateau**

 **FC exotherm bump**

 **Maillard valley**

 **Free-fall (exp)**

☐  **Precharge start**

☐  **Coast to drop**

 **Sprint through the lows**

race to °C by min sec — then dwell in that band until dry end

Pick a shape — the curve is built to hit your dry-end, first-crack and drop targets at exactly those times, and loads into the graph below. Flip between the shapes to compare, tweak points if you like, then **Save profile**. **Declining**: RoR starts high, eases down all roast (classic). **Pyramid**: climbs to a peak at dry end, then declines. **Straight**: constant RoR per phase. **Soak**: near-flat first ~45 s, then declining. **Flat development**: declining until FC, then RoR held steady. **Step-down at FC**: sheds RoR ~30 s before FC (anti-crash/flick), calm flat development. **Boil-off plateau**: parks ~60-90 s at the boiling point so energy evaporates moisture at constant temp, then declines. **FC exotherm bump**: a small deliberate RoR *rise* into FC so the PID holds power through the steam crash — the real RoR comes out flat. **Maillard valley**: hot drying into a near-stall dwell past dry end (time-at-temperature for sweetness), second push, decline — experimental. **Free-fall**: exponential RoR decay, the natural curve of a constant heater. **Sprint**: flat out to the sprint temp (140 °C by default) by the time you set, then a slow, steady dwell through the band from there to dry end — the bulk of drying is spent in that band, and the RoR deliberately picks up again into Maillard (a valley, not a decline; the heater will be at full power for the sprint). Modifiers combine with any shape —  **Precharge** commands +8 °C for the first 40 s (full power into cold wet beans, faster turnaround);  **Coast** ends the command 3 °C early and lets stored drum heat carry the beans to the drop. The fan is not set — tick **Auto fan** below or fill the fan column.

Fig. 11 — The generator: phase temperatures and durations, the shape picker, and the two modifiers.

Hands-on: draw on the graph

- Press  Graph locked — tap to edit to unlock.
- Drag points to move them, press empty space to add a point, double-tap a point to delete it. The implied RoR of each segment is shown while you edit.

- The **phase bar** uses the **DE at** and **FC at** temperatures beside it. Set those to match your beans.
- Fine-tune numbers in **Details — points table**.

▼ Details — points table

TIME M:SS	TARGET °C	FAN % (OPT)	
0:00	100	opt	×
1:00	124	opt	×
2:00	146	opt	×
3:00	163	opt	×
4:00	177	opt	×
5:00	188	opt	×
6:00	196	opt	×
7:00	202	opt	×
8:00	206	opt	×
8:30	208	opt	×

Fig. 12 — The points table: time from Charge, target °C and an optional per-point fan %.

Fan

- **Auto fan** (recommended): the fan follows the *measured* bean temperature, declining from the start % to the end %. An optional **mid** point lets you shape a rise-then-fall. This overrides per-point fan values while running.
- Or fill the **Fan %** column per point for explicit control.
- Or leave both empty to run the fan by hand during the roast.

Import and export

Export JSON

saves the profile as a file you can share or hand to an AI assistant to tweak.

Import JSON

loads one. The format is documented in `PROFILE_FORMAT.md` in the repository. Paste that file into Claude or another assistant to have it design a curve for a specific bean.

Running a profile without the between-batch cycle

Select it in the drop-down and press

▶ Run profile

. If Charge was marked within the last 10 seconds the charge soak applies. Otherwise the PID engages on the curve immediately, which is useful for restarting mid-roast.

7. Reading the screen during a roast

Live tiles

Bean (BT), Exhaust (ET), RoR (rate of rise in °C/min), Target (the setpoint the app is commanding), Heater % and Fan %. The **PID** pill shows whether the roaster's closed-loop mode is engaged. Below the tiles: the previous event, a live phase bar, and the estimated time to first crack.

Chart

Top pane: temperatures on the left axis, RoR on the right (0–30 °C/min). Bottom pane: heater and fan %, with the dashed gold line showing cumulative heat. Dashed curves are the plan. Faded curves are a background roast (section 8). Hover or drag on the chart to read values.

Under the chart

- The **Enable control** tick.
- When you have adjusted the fan by hand during a profile, the Resume profile fan and Resume — shift curve to my fan buttons appear here.
- **RoR window** and **RoR smooth** tune how responsive versus how calm the RoR trace is.

Control rail

On wide screens a **Controls** column sits to the right of the chart (on phones it appears below it), laid out like the stock app's side panel: a **Target temp** tile and a **PID** tile, **Heater –/+** and **Fan –/+** steppers showing the live values, a green **event-tagging** button that tags **DE** then **FC**, and a red **Stop / Drop**. Both steppers move 1 % per tap. This is where you mark events, adjust the fan and end the roast. Heater ± takes **manual control**: it switches the PID off and stops any running profile or cycle (you are asked to confirm), then sets the heater directly. Stop ends the roast: heater off, fan back to the charge speed, roast saved.

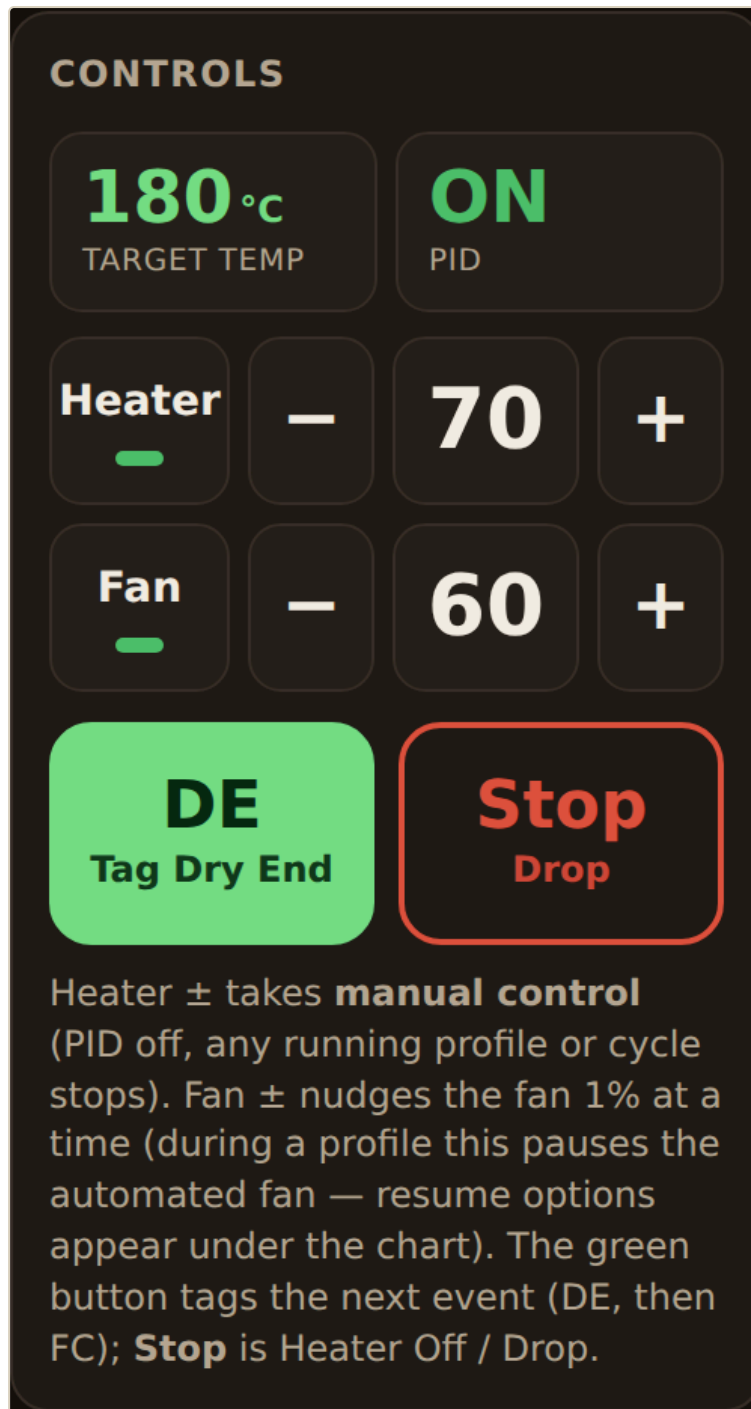


Fig. 13a — The control rail during a roast: the next event to tag is Dry End.

Manual mode

If you would rather roast by hand than follow a profile, choose  **Manual** under **At bean drop** in the **Between batches / charge** panel. The cycle cools, charges and lights **DROP BEANS NOW** exactly as before. When the beans go in, the app marks Charge, starts the timer and batch code, runs the charge soak if it is on, then sets your **opening heater %** and **fan %** with the PID **off** and hands over. From there you drive the roast with the Heater and Fan ± steppers, tag DE and FC with the green button, and press Stop / Drop to end and save. The opening values are remembered between sessions.

BETWEEN BATCHES / CHARGE

Start between-batch cycle

Cools the machine (fan 100%, heater off) to the cool-to temp, then ramps gradually to your charge temp over **1 min** at the charge fan speed, holds it **30 s**, and lights up when it's time to drop the beans.

At bean drop ☐ **Run the profile** (or an armed replay) ☒ **Manual** — hand control from the Controls rail

Opening heater % · fan

% — set at Charge (after the charge soak if it is on), PID stays **off**; from there you drive Heater / Fan ± yourself. Stop / Drop ends and saves the roast as usual.

At bean drop this will run:

Manual — hand control

Charge is marked, then the charge soak runs (30% to 0:30, 40% to 0:45), then heater 55% · fan 65% · PID off — you drive from the Controls rail

Cool to °C	Charge temp °C	Charge fan %
100	180	30

☒ **Charge soak** — heater % until s, then % until s

Fixed heater power right after **Charge** (open loop, PID off) — then the control loop takes over. Applies to profile runs **and ET-track runs** started at charge. **Untick to skip the soak** — the control loop then engages on the curve from the first second after charge. Saved in this browser. ⚡ Charging and the between-batch cycle are **hard-capped at 85% power** — above that the app pauses the PID and holds 85% until the temperature is nearly there.

☒ **Auto-charge** — when the beans drop (BT plunges), mark Charge and start the profile loaded in the designer automatically. **No need to press ► Run profile** — the whole roast runs from this one button.

Fig. 13b — Manual mode selected: opening heater 55 %, fan 65 %. The preview box spells out what will happen at the bean drop.

Live guidance

A dotted 60 s projection of where the bean temperature is heading, an FC countdown, and automatic **RoR crash / flick** warnings appear as the roast progresses.

Fan override

Any manual fan change during a profile pauses the profile's fan while the temperature curve keeps running.

hands it back at the profile's values. hands it back but offsets the whole remaining fan curve through the speed you chose.

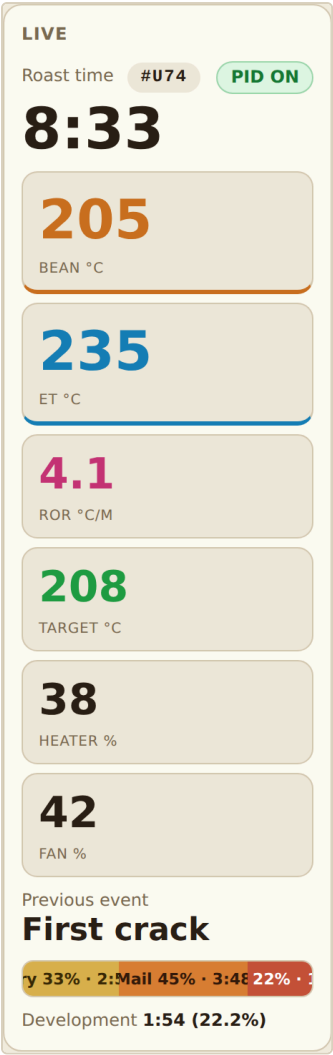


Fig. 13 — Late in the roast: First Crack is the previous event and the phase bar shows development.

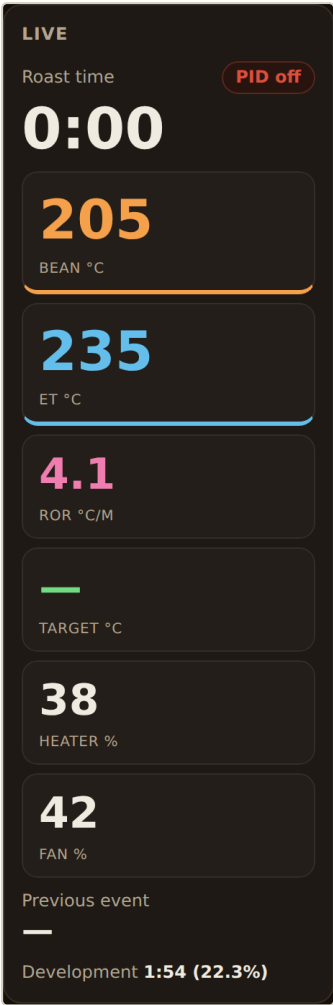


Fig. 14 — The same rail in night mode.

8. After the roast: data, notes and comparisons

Everything is in the **Data** panel.

- Roasts are **saved automatically at Drop** with a date and time stamp and their batch code. **Save this roast** stores the current chart manually, for example after a test run. The drop-down lists saved roasts for **Load** and **Del**.
- **Roast notes** follows the current roast (the one roasting, just dropped, or loaded from the list) and saves as you type. **☆ Star** marks an excellent roast so it stands out in the list.
- The bean and green weight you chose in the Start cycle dialog are stored with each roast and shown in its list entry.
- **Download CSV** exports the current chart's samples (time, bean and exhaust temperatures, heater, fan, target, events) with the roast details in a header, for Excel or Artisan.
- **Overlay as background** ghosts a saved roast onto the chart, aligned at Charge, and shows a live **BT delta vs background** readout so you can roast to match it. **× clear BG** removes it.

DATA

Download CSV **Clear chart** 256 samples · #U74 Ethiopia Yirgacheffe, washed

Save this roast ★ #U74 · Sep 4 09:44 AM · Ethiopia Yirgacheffe, washed **Load** **Del**

Roast notes — #U74 Ethiopia Yirgacheffe, washed **★ Starred**

Sweet, clean cup. Slightly faster drying next time.

Overlay as background

Background overlays a past roast (ghost curves, aligned at Charge) on the live chart so you can roast to match it — the Live panel shows your BT delta against it.
Saved roasts live in this browser. Load one to review its chart; Download CSV exports whatever's on the chart now.
► Raw fields & log (for verifying the ET channel)

Fig. 15 — The Data panel with a starred roast loaded and a note typed in.

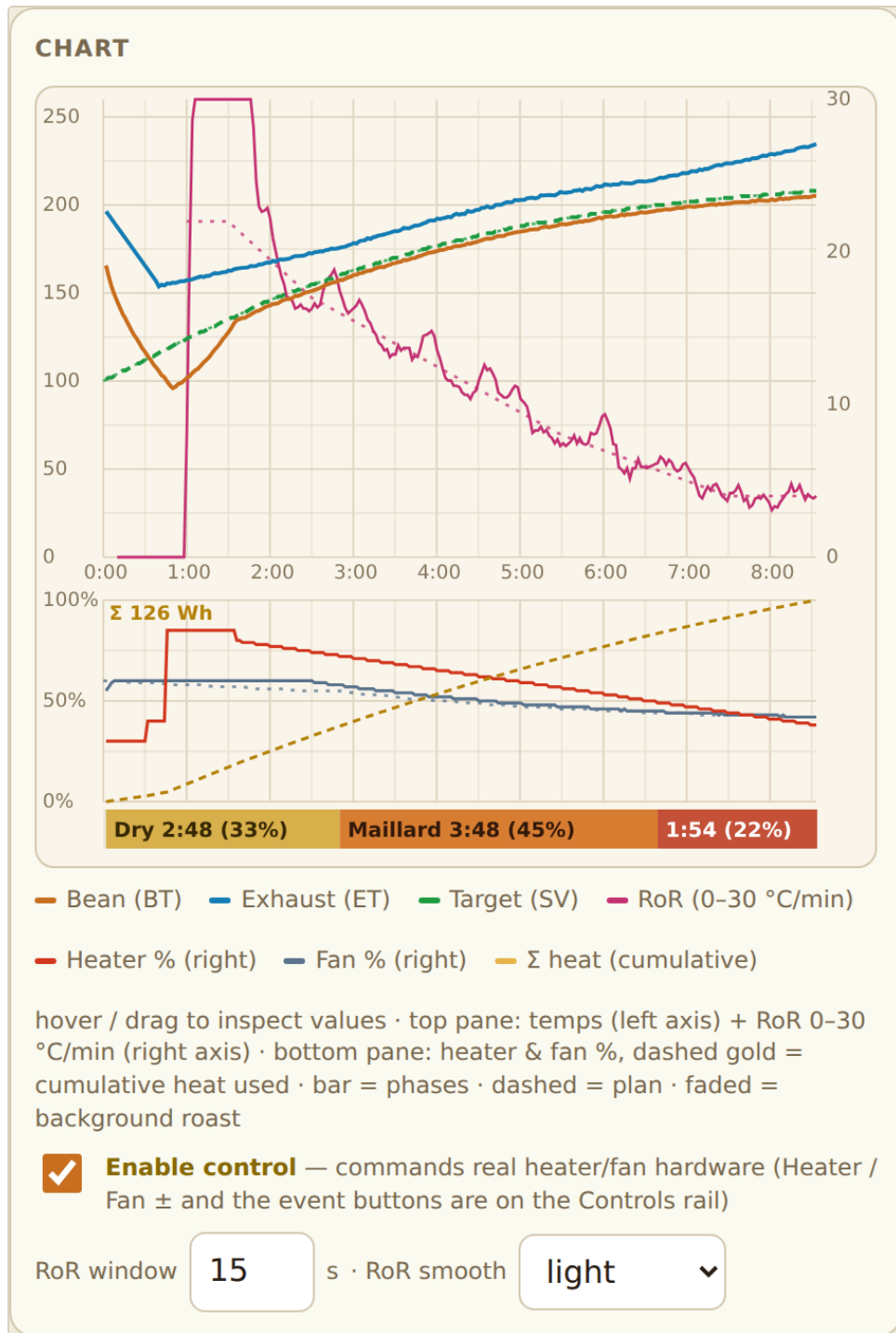


Fig. 16 — A saved roast loaded for review. Use Overlay as background to ghost it behind your next live roast.

9. Green coffee inventory

Optional, but it makes batch planning and stock tracking automatic.

1. In **Green coffee inventory** press **+ Add bean** and enter the name, origin, available weight and notes. Or press **Photo label (OCR)** and photograph the bag's spec label. On-device text recognition (English and Chinese) fills the fields for you to check.
2. Set your roaster's **Batch size** range (for example 100–200 g). Each bean card then shows a suggested even split, such as $5 \times 170\text{ g}$, so you do not end up with an awkward leftover. **Batch planner** does the same for any weight you type in.
3. When starting a roast, pick the bean and the green weight in the dialog that opens from **Start between-batch cycle**.
4. At **Stop / Drop** the batch's green weight is deducted from stock. Out-of-stock beans are flagged in the list.

GREEN COFFEE INVENTORY

+ Add bean

Photo label (OCR)

▶ AI label reading (optional)

Batch size

100

 -

200

 g per roast — used to plan splits below.

Ethiopia Yirgacheffe · Washed, crop 2025

Available

Batches

Floral, citrus. Filter roast around 205 °C.

Edit

Dup

Del

850.0 g

5 × 170 g

Brazil Cerrado · Natural, 2025

Available

Batches

Chocolate, nutty. Good espresso base.

Edit

Dup

Del

420.0 g

3 × 140 g

▶ Batch planner

Fig. 17 — Two beans in stock, each with its suggested batch split.

Inventory item

Bean name

Colombia Huila

Origin / farm

Washed, crop 2025

Available g

1000

Notes

Caramel, red apple.

Cancel

Save

Fig. 18 — Adding a bean by hand.

GREEN COFFEE INVENTORY

+ Add bean

Photo label (OCR)

▶ AI label reading (optional)

Batch size

100

 -

200

 g per roast — used to plan splits below.

Ethiopia Yirgacheffe · Washed, crop 2025

Available

Batches

Floral, citrus. Filter roast around 205 °C.

Edit

Dup

Del

700.0 g

4 × 175 g

Brazil Cerrado · Natural, 2025

Available

Batches

Chocolate, nutty. Good espresso base.

Edit

Dup

Del

420.0 g

3 × 140 g

▶ Batch planner

Fig. 19 — After the roast: 150 g deducted from the Ethiopia, and the split re-planned.

Batch planner

Green to split

Ethiopia Yirgacheffe · 850 g

▼

850

✓

5 × 170 g

— even split, nothing left over.

vs max-size: 4 × 200 g + 50 g left over (too small to roast).

Other even splits: 6 × 141.7 g · 7 × 121.4 g · 8 × 106.3 g

Fig. 20 — The batch planner splits any weight into equal batches within your size range.

AI label reading (optional)

Inside  **AI label reading** you can point the photo reader at Claude or Gemini for much better results on messy or multi-language labels. Paste **your own** API key (stored in this browser only and never synced), or set a **Proxy Worker URL** so the key never lives on the device. The photo is sent to your own account and billed to you. Leave both fields empty to keep using on-device OCR.

AI label reading (optional)

Provider

Claude (Anthropic) ▼

API key

paste your key (stored in this browser only)

Proxy Worker URL

optional — https://...workers.dev (keeps the key off this device)

With a key set,  **Photo label** sends the photo to your chosen AI to read the label — far better on messy, dense, or multi-language labels. It fills the name, origin, bag weight, and a tidy spec summary. The photo is sent to **your own account** (billed to you). Get a key at **console.anthropic.com** (Claude) or **aistudio.google.com** (Gemini — has a free tier). Leave everything blank to use on-device OCR instead.

 **Proxy Worker URL** (optional): route through your own Cloudflare Worker (deploy **ai-proxy-worker.js** from the repo) so the key lives on the Worker, not this browser — best for shared/public use. When a proxy is set the API-key field above is ignored. The key is kept only in this browser and never cloud-synced.

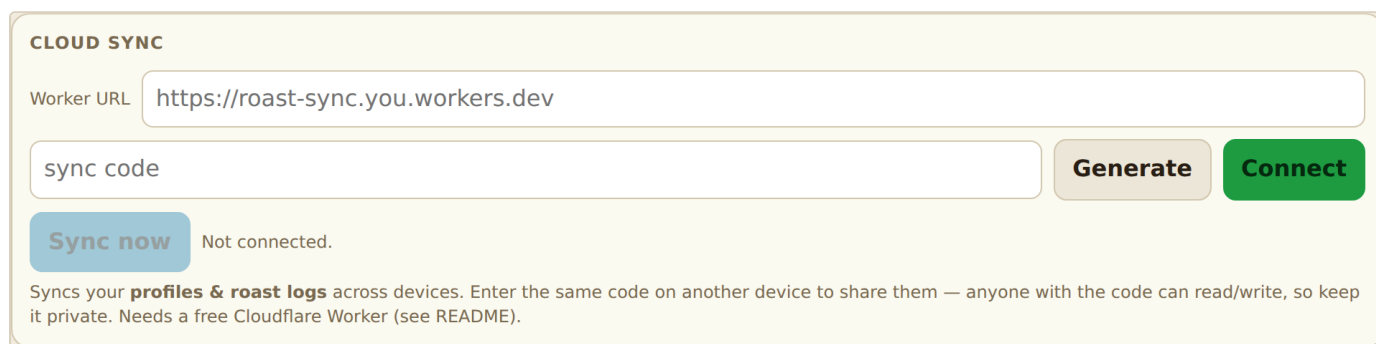
Fig. 21 — Provider, API key and optional proxy Worker URL.

10. Using more than one device (Cloud sync)

Profiles, roast logs and inventory can be shared between your phone, tablet and computer through a small Cloudflare Worker that **you** own. It is free and needs no account in the app.

Setup takes about 10 minutes once and is walked through click by click in `CLOUD_SYNC.md` in the repository. In short:

1. Create a free Cloudflare Worker from `sync-worker.js` with a KV namespace bound as `SYNC`.
2. Paste the Worker's URL into the **Cloud sync** panel.
3. Press **Generate** for a sync code, then **Connect**. Your local data is uploaded.
4. On every other device enter the same URL and code and press Connect. From then on syncing is automatic: a change pushes about 1.5 s after you make it, each device pulls every 45 s and whenever you switch back to the tab.



The screenshot shows the 'CLOUD SYNC' panel. At the top, it has the title 'CLOUD SYNC'. Below it is a 'Worker URL' input field containing 'https://roast-sync.you.workers.dev'. Underneath is a 'sync code' input field. To the right of the 'sync code' field are two buttons: 'Generate' (light blue) and 'Connect' (green). Below the 'sync code' field is a blue button labeled 'Sync now' followed by the text 'Not connected.'. At the bottom of the panel, there is a small text block: 'Syncs your **profiles & roast logs** across devices. Enter the same code on another device to share them — anyone with the code can read/write, so keep it private. Needs a free Cloudflare Worker (see README).'

Fig. 22 — The Cloud sync panel.

The sync code is both the address and the password. Anyone with it can read and write your data, so keep it private. Do not share your Worker URL publicly either.

11. Advanced mode (experienced users only)

Ticking **Advanced mode** below the Profile designer reveals two open-loop tools that drive the heater **directly in manual mode**. Stay at the machine while they run. **Stop / Drop** still works.

- **ET-track roast** — replays a saved roast by its *exhaust* temperature, using the reference's recorded heater % as feedforward with trim on the ET error, and replays its fan curve. The run ends when the bean temperature reaches the **Drop at BT** you set. Arm it for auto-charge to use it in the between-batch flow instead of a profile.
- **Power profile** — distils a saved roast's heater curve into a fixed schedule of power steps (one **time** heater% [fan%] per line) and runs it with zero mid-roast adjustments. Set **Drop at BT** as a safety net.

Both are experimental. Do a few normal profile roasts first so you know how your machine behaves.

ET-TRACK ROAST (EXPERIMENTAL)

⚠ Drives the heater **directly in manual mode** (firmware PID off) with a software control loop. Stay at the machine — **Heater Off / Drop** always works.

— pick a reference roast —

Smooth: medium (20 s)

Drop at BT

°C — the run ends (heater off) when the bean temp reaches this, not on the reference's clock.

☐

Arm for auto-charge — the bean-drop detection starts this ET-track run instead of the normal profile, and the reference shows as the chart background.

▶ Start ET-track run

Replays a captured roast by its **exhaust temp**: pick a saved roast, and the app follows its ET curve — feedforward from the reference's recorded heater % plus PID trim on the ET error — while replaying its fan. **Smooth** cleans the captured line before it becomes the target — heavier smoothing means fewer heat adjustments. The target is cut at the reference's drop point and keeps its true end temp, however heavy the smoothing.

▶ Control tuning (advanced)

Fig. 23 — ET-track roast.

POWER PROFILE (OPEN LOOP)

⚠ Drives the heater **directly in manual mode** on a fixed schedule — no temperature feedback at all. Stay at the machine — **Heater Off / Drop** always works.

— pick a source roast —
step ≥ 2 %
Generate

```

one step per line:  time  heater%  [fan%]
0:00 30 60
0:45 78
3:10 74
...
9:40 0  ← last line: heater off (end of schedule)

```

Drop at BT °C — safety net: the run ends (heater off) when the bean temp reaches this, even mid-schedule.

☐ Arm for auto-charge — the bean-drop detection runs this power schedule (takes priority over ET-track and the normal profile).

▶ Run power profile

Generate distills a saved roast's recorded heater curve into fixed power steps: the roast then runs purely on this schedule — your charge soak first, then the steps, **zero mid-roast adjustments**. The generated schedule rises once to its peak, then **only ever lowers the heat** — wobbles are **averaged, not clipped**, so it delivers the **same total heat ($\Sigma \text{ \%} \cdot \text{min}$)** as the source; the line under the plot shows the comparison (live as you edit). Edit the lines freely; the schedule is saved in this browser.

Fig. 24 — Power profile (open loop).

12. Safety checklist

- Fan on before heater on. The app enforces airflow in its automated flows, but in manual fan control it is on you.
- **Enable control** is per session. It resets when you reload the page, which is intentional.
- Stay with the machine. Alarms and warnings are aids, not supervision.
- Losing the Bluetooth link stops every automated run and the app shows **disconnected**, but the roaster's own state persists. If a roast is in progress when the link drops, reconnect immediately or turn the roaster's heater off at the machine.
- If in doubt: Stop / Drop on the Controls rail.

13. Troubleshooting & FAQ

Problem	What to try
"Web Bluetooth isn't available" banner	Use Chrome or Edge, not Safari or Firefox. On iOS use another device.
Roaster not in the device list	Power-cycle the roaster, close the vendor app, tick List all Bluetooth devices , move closer.
Connected but tiles show 	Open Connection details / advanced . If the Notify/Write fields are  , the service was not found: use the free nRF Connect app to read the roaster's Service UUID, paste it into the custom Service UUID box, then reconnect.
Buttons do nothing	Enable control is not ticked, or the roaster is disconnected.
Heater will not fire	The fan is at 0. Set a fan speed first.
Auto-charge did not trigger	The bean-probe drop was too small or slow (small batch, low charge temperature). Press ▶ Run profile to start the curve now. Next time use a larger batch or a higher charge temperature so the plunge is clearer.
RoR trace is jagged	Increase RoR smooth or lengthen RoR window under the chart.
Profile started but the fan stayed put	No fan values in the profile and Auto fan off. Control the fan by hand, or tick Auto fan and re-run.
My saved data is gone	Data is stored per browser and per site URL. Check you are on the same URL and browser, or set up Cloud sync. Clearing site data deletes it.
I want °F	Not supported yet. The app works in °C throughout.

Still stuck? Open **Raw fields & log** at the bottom of the Data panel. It shows the raw telemetry frames and a log of every command sent, which is the first thing to include when asking for help.